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Frontend Assignment

→ Basic Objective

Create a redux based counter app:

- Initial app : 0 (value)
- Two buttons
  - Increment (+)
  - Decrement (-)
- Count update based upon button clicks.

→ Concepts used:

- Redux Toolkit (@ reduxjs/toolkit)
- React-Redux (use Selector, use Dispatch)
- Centralized state managmnt.

→ Install Dependencies

npm i @ reduxjs/toolkit react-redux

→ Create redux store (store.js)

import {configureStore} from "@ reduxjs/toolkit";

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```
import counterReducer from "./counterSlice";
export const store = configureStore ({
  reducer: {
      counter: counterReducer;
  },
    });
```

→ Create counterSlice.jsx

```
import {createSlice} from "@reduxjs/toolkit";

const counterSlice = ~~Counter~~ createSlice ({
  name: "counter"
initialstate : { value 0 },
  reducers : {
    increment : (state) => {
      state.value += 1;
      },
    decrement : (state) => {
    state. value -= 1;
      },
      },
      });
export const { increment, decrement } =
    counterslice. actions;
export default counterSlice.reducer;
```

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→ Create App.jsx (update)

```
import {useSelector, useDispatch } from
    "react-redux"
import {increment, decrement } from
    "./redux/counterSlice";

const App = () => {
    const count = useSelector((state)
    => state.counter.value);
    const dispatch = useDispatch();

return (
<div>
    <h1> Redux Counter </h1>
    <h2> {count} </h2>
<button onClick = {() =>
    dispatch (increment ())}>
    Increment
    </button>

<button onClick = {() => dispatch
    (decrement ())} >
    Decrement  </button>
</div>
);  };  export default App;
```

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→ Update main.jsx

```
import ReactDOM from "react-dom/client";
import App from "./App";
import {Provider} from "react-redux";
import store from "./redux/store";

ReactDOM.createRoot(document.getElementById
  ("root")).render(
  <React StrictMode>
   <Provider store={store}>
      <App />
      </Provider>
    </ React.strictmode> );
```

→ Folder Structure

```
src/
 ├→ App.jsx
 ├→ main.jsx
 └→ redux/
      ├→ store.jsx
      └→ counterSlice.jsx
```